DEMO RUN INSTRUCTIONS

Run the MIRQAB Demo

A localhost-only evaluator demo, started with one command and populated automatically with public, license-cleared security data.

TEAM

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PROJECT

MIRQAB — AI-Powered Security Operations Platform for SMBs

27 September 2026

At a glance

What you get	A self-contained MIRQAB stack (PostgreSQL, Redis, backend, SOC Dashboard and gateway) with the APT29 and PcapMonkey demo data imported through MIRQAB's own pipeline
URL	`http://localhost:8080` (localhost only)
Login	The admin email and password **you** choose in `infra/demo/.env.demo`
Safe by design	Cannot isolate any endpoint, uses no external AI, publishes PostgreSQL/Redis nowhere, and uses its own Compose project and volumes (`mirqab-demo`)
Commands source	Every command below is taken from the project's own `infra/demo/README.md` and `docker-compose.demo.yml`

Verified end-to-end (27 Sep 2026)

These exact instructions were followed from the packaged source zip on the development machine. The one command finished successfully, all 5 services ran, and 48 of 48 checks passed: API, RBAC, realtime, reports, a browser walkthrough and the response window. The demo was then removed cleanly.

Not yet verified

The demo has **not been run on a separate clean machine**. On the development machine, Docker and the base images were already available. If something differs on your machine, see Troubleshooting.

1. Requirements

- **Docker Desktop** (Windows/macOS) or **Docker Engine** (Linux) with **Docker Compose v2** (`docker compose version`)
- **Internet access for the first start**: Docker pulls the base images (postgres:16-alpine, redis:7-alpine, node:20-alpine, nginx:1.25-alpine) and the image builds install npm packages
- Free local port **8080** (or choose another with `DEMO_HTTP_PORT`)
- A modern web browser

2. Get the project

Unzip `12_Project_Source/MIRQAB_Source_Code.zip` from this package and open a terminal in the extracted `MIRQAB-source` folder, the repository root. (If the team shares a Git repository link, cloning it is equivalent.)

3. Configure `.env.demo` from the example

```
# from the repository root
cp infra/demo/.env.demo.example infra/demo/.env.demo
# Windows PowerShell alternative:
Copy-Item infra/demo/.env.demo.example infra/demo/.env.demo
```

Edit `infra/demo/.env.demo` and fill in:

Variable	What to put
`DEMO_ADMIN_EMAIL`	Admin login email (default `admin@mirqab.local`)
`DEMO_ADMIN_PASSWORD`	Your own admin password (8+ characters). Required.
`POSTGRES_PASSWORD`, `JWT_ACCESS_SECRET`, `JWT_REFRESH_SECRET`, `MONITORING_INGEST_API_KEY`	Random strings (JWT secrets at least 32 characters). Required. Never reuse real values.
`DEMO_ANALYST_*`, `DEMO_TECHNICIAN_*`	Optional. Set both email and password to also create a Security Analyst and an IT Technician (the technician sees only the APT29 demo department), which demonstrates RBAC.
`DEMO_HTTP_PORT`	Optional (default 8080)

Generate random secrets with either:

```
openssl rand -hex 32
# PowerShell:
-join ((1..64) | % { '{0:x}' -f (Get-Random -Max 16) })
```

4. One-command setup

```
docker compose -f infra/demo/docker-compose.demo.yml --env-file infra/demo/.env.demo up -d --build
```

The first start takes several minutes because it builds the images. The data import itself takes a few seconds once the backend is up. On start, the backend applies database migrations, creates the demo organization, admin, departments, devices and policies (only if missing), and imports the bundled datasets (each event at most once). Running the command again is safe: it never resets or deletes data.

5. Health check

```
# all five services should be "running" (postgres and redis "healthy")
docker compose -f infra/demo/docker-compose.demo.yml --env-file infra/demo/.env.demo ps

# backend start-up log: look for "Demo data ready" and "Nest application successfully started"
docker compose -f infra/demo/docker-compose.demo.yml --env-file infra/demo/.env.demo logs backend
```

Expected log lines (from the verification run):

```
All migrations have been successfully applied.
Seeded initial admin user: admin@mirqab.local
[DemoDataService] Demo data ready: {... "apt29":{"total":22,"imported":22,...},"pcap":
{"total":6,"imported":6,...}}
[NestApplication] Nest application successfully started
```

Warnings saying that external AI, Wazuh and endpoint control are disabled are **expected** in the demo.

6. Demo login and local URL

1. Open `http://localhost:8080` (or your `DEMO_HTTP_PORT`).
2. Sign in with `DEMO_ADMIN_EMAIL` / `DEMO_ADMIN_PASSWORD`. The interface opens in Arabic; use the language button (top right) to switch to English.

7. Demo workflow (about 10 minutes)

#	Step	What to do / expect in the demo
1	Dashboard	Overview: KPIs, live network map and active incidents built from the imported datasets (27 incidents in the verification run).
2	Incidents	Sidebar → Incidents. Filter by **High**. Incidents are labelled "Open-Source Dataset".
3	Investigation	Open a HIGH incident: Response and Investigation windows, with what happened and attack progression.
4	Evidence	Expand the evidence (*Read more*) to see the original dataset text.
5	MITRE	Security Analysis shows the tactic and technique. Some incidents are honestly *Unclassified*.
6	Suggested Solutions	Similar incidents and recommended fixes. A fresh demo has no response history, so "effective solutions" may be empty.
7	Response timer	HIGH incidents get a 5-minute window from detection. Once it expires unacknowledged, the demo shows "automated containment unavailable: no endpoint-control provider configured". Try **Acknowledge & Investigate** on one.
8	Network map	Live Network Map and The Citadel → Attack Path (APT29 plan).

#	Step	What to do / expect in the demo
9	Isolation workflow	Isolate Endpoint shows the confirmation dialog. No endpoint-control provider is configured, so **no device can be isolated**. The request stays PENDING rather than claiming success (observed in the verification run).
10	Notifications	The bell and the Notifications page show "response required" and "containment result" entries.
11	Reports	Download Case Report (PDF) from an incident. Reports page: generate the weekly report.
12	RBAC (optional)	If you created the technician account, sign in as the technician: only the APT29 demo department is visible.

8. Safe shutdown

```
# stop the demo (keeps its data for next time)
docker compose -f infra/demo/docker-compose.demo.yml --env-file infra/demo/.env.demo stop
```

Start it again later with the same `up -d` command. Do **not** add `-v` to a `down` command unless you intentionally want to delete the demo database. The demo never touches any other MIRQAB database or volume.

9. Limitations to be aware of

- Verified end-to-end on the development machine only; not yet run on a separate clean machine.
- Localhost only, plain HTTP (no TLS) on `127.0.0.1`. It is not intended for network exposure.
- No endpoint control and no external AI: isolation and AI-model features cannot be demonstrated end-to-end in the demo. See the Security and Prototype documents for the lab evidence.
- Detection covers 12 MITRE techniques, so part of the dataset stays *Unclassified*.
- First build needs internet access and takes several minutes.

10. Troubleshooting

Symptom	Likely cause / fix
Compose error "Set … in infra/demo/.env.demo"	A required value is empty. Fill it in and rerun.
Port 8080 already in use	Set `DEMO_HTTP_PORT` (for example 8081) and open that port instead.
Browser shows an error right after start	The backend is still starting or importing. Wait and check the backend log for "successfully started".
Login fails	Use exactly the email and password from `.env.demo`. After 5 failed attempts within a minute, wait one minute.
Backend restarts repeatedly	Read `logs backend`. A migration or configuration error is printed there.